Exploring uncertainty in Antarctica's future sea level contribution

Jonathan Barnsley

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Total word count: 8215 words.

Chapter 1

Literature Review

1.1 Sea level change

509 words

The IPCC's Sixth Assessment Report (AR6) identified sea level rise as "an existential threat for island nations, low-lying coastal zones, and the communities, infrastructure, and cultural heritage therein" (Cooley et al., 2022). A rise of 1 m would increase the global population living below mean sea level from between 34–49 million people today to 77–132 million (Seeger and Minderhoud, 2026). Sea level also fluctuates about its mean state due to tides, storm surges, and other variability (Woodworth et al., 2019). Peaks in this variability coincide to produce extreme sea level events, which are made more frequent and severe by rises in mean sea level (Boumis et al., 2023). The impacts of sea level rise are therefore not only felt in a binary way (i.e. land is either above or below sea level), but rather as a continuously increasing risk to which communities are exposed. This exposure is highest in the Global South, meaning that some of the communities most at risk are also the least economically and institutionally equipped to adapt (Seeger and Minderhoud, 2026; O'Neill et al., 2022).

Due to anthropogenic climate change, global mean sea level rise has accelerated from 1.35 ± 0.57 mm yr^{-1} between 1901–1990 to 3.69 ± 0.48 mm yr^{-1} between 2006–2018 (Fox-Kemper et al., 2021). AR6 projects a *likely* sea level rise of 0.33–0.62 m by 2100 under a low emissions pathway and 0.63–1.01 m under a high emissions pathway, the upper bound of which coincides with the 1 m sea level rise hypothetical evaluated by Seeger and Minderhoud (2026). Global mean sea level rise occurs either through thermosteric effects ("thermal expansion") or through barystatic contributions from land water storage or land ice (Gregory et al., 2019). The dominant drivers of historic sea level rise have been thermal expansion and melting mountain glaciers. However, contributions from the polar ice sheets have accelerated since the 1970s and are projected to be significant sources of sea level rise this century (Fox-Kemper et al., 2021). Antarctica is the largest source of uncertainty in these projections, due in large part to poorly constrained processes governing the ice sheet response (Edwards et al., 2021).

Beyond 2100, AR6 cites "deep uncertainty" in multi-century sea level projections. The 21st century projections are informed by two model intercomparison projects, which combine the efforts of numerous modelling groups to explore uncertainty in the ice sheet contribution to future sea level rise (Levermann et al., 2020; Payne et al., 2021). However, no such project existed for multi-century projections at the time. Instead, AR6 combined information from multiple sources, including extrapolation of the 2100 rates of mass loss, expert elicitation, and the existing body of literature. Under high emissions, It projected 1.7–6.8 m of sea level rise, but could not rule out sea level rise of up to 15 m arising from possible ice sheet instabilities (DeConto et al., 2021). Turner et al. (2023) give probabilistic projections of multi-century sea level rise in low and medium emissions scenarios: 0.83–3.38 m and 1.72–5.85 m, respectively. They again cite the Antarctic Ice Sheet as the largest source of uncertainty and underline the need for multi-century model ensemble projections. Significant progress has been made in this area over the last few years (see Section 1.4.3), but constraining the uncertainty in Antarctica's sea level contribution remains a challenge, thus demonstrating the need for further work.

1.2 The Antarctic Ice Sheet

1.2.1 Geographical setting

537 words

The Antarctic Ice Sheet is the largest store of freshwater on Earth, holding 26.0 ± 0.4 million km^3 of ice (Morlighem et al., 2020). It is a dynamic ice mass, accumulating ice through precipitation and flowing outwards under gravity. This flow organises into ice streams, which are fast-flowing corridors of ice that transport ice from the ice sheet interior to its perimeter (Morlighem, 2020). 93% of the ice sheet perimeter interfaces with the ocean, either as a floating ice shelf (74%) or vertical ice cliff (19%) (Bindschadler et al., 2011). Ice shelves lose mass through a combination of iceberg calving and basal melting (Davison et al., 2023). The point of transition between grounded ice sheet and floating ice shelf is called the "grounding line" (Friedl et al., 2020).

Antarctica can be split into three climatically distinct regions: West Antarctica, East Antarctica, and the Antarctic Peninsula. The West Antarctic Ice Sheet (WAIS) is grounded on bedrock that lies predominantly below sea level (a marine ice sheet), which could make it susceptible to instabilities driving self-sustaining mass loss (see Section 1.3.2). This is particularly concerning because West Antarctica is the region with the highest current rate of mass loss (Table 1.1) and may have collapsed completely in past warm climates (see Section 1.2.3). West Antarctic ice streams drain into the Amundsen Sea, Ross Sea, and Weddell Sea sectors of the Southern Ocean. The latter two feature Antarctica's two largest ice shelves, the Ross ice shelf (**TODO: value** km^2) and the Filchner-Ronne ice shelf (**TODO: value** km^2) **TODO: cite**.

Table 1.1: Antarctica's component ice sheets, their rate of mass loss between 1992–2020 (Otosaka et al., 2023), and their total sea level potential (Morlighem et al., 2020).

Region	Mass loss (Gt yr^{-1})	Sea level potential (m)
WAIS	-82 ± 9	5.3 ± 0.2
EAIS	3 ± 15	52.2 ± 0.7
APIS	-13 ± 5	0.27 ± 0.02

The East Antarctic Ice Sheet (EAIS) is separated from West Antarctica by the Trans-Antarctic mountains. It represents by far the largest share of Antarctica's sea level potential, but has contributed the least to recent sea level change (Table 1.1). Despite the current trends, the EAIS may have been more dynamic in past climates and its future response is highly uncertain (Stokes et al. 2022). Although the EAIS is mostly terrestrial (grounded above sea level), its underlying bedrock has three key marine basins: The Recovery, Wilkes, and Aurora subglacial basins. These correspond with the Coats Land, George V Land, and Wilkes Land sectors of Antarctica, respectively. The Totten glacier, which drains most of the Aurora subglacial basin, alone holds 3.89 m sea level equivalent of ice (Morlighem et al., 2020). These basins may be susceptible to the same instabilities as West Antarctica if they are sufficiently pushed from their current state (Mengel and Levermann, 2014; Aitken et al., 2016; Golledge et al., 2017a).

The Antarctic Peninsula Ice Sheet (APIS) is a comparatively small part of Antarctica, but is losing mass faster than any other region relative to its size (Table 1.1). It is distinct as the most Northerly region of the continent, reaching 63°S towards the Drake Passage. Its mountain range features smaller, Alpine-like glaciers down steep terrain. The Peninsula acts as an orographic barrier between the Weddell Sea and the Bellingshausen Sea, strongly influencing the climate on either side (Davies et al., 2026). The largest ice shelf in the APIS, the Larsen ice shelf, extends down the Eastern flank of the peninsula and is divided into smaller sections Larsen A to D, the largest of which is Larsen C at **TODO: value and cite** km^2. The recent changes in the Antarctic Peninsula are significant not only to the local environment, but also serve as a bellwether for processes that may later occur in East and West Antarctica with substantially greater consequences for sea level.

1.2.2 Recent observed changes

Antarctica is not in equilibrium with the current climate, with some areas showing a clear trend in mass balance that is consistent across multiple methods of measurement (e.g. Gardner et al. 2018; Smith et al. 2020; Velicogna et al. 2020). Furthermore, mass loss from West Antarctica has accelerated across every four year period from -37 ± 19 Gt yr^{-1} in 1992–1996 to -131 ± 21 Gt yr^{-1} in 2012–2016, before slowing to -94 ± 25 Gt yr^{-1} in 2017–2020 (Otosaka et al., 2023). The vast majority of this mass loss has originated from the Amundsen Sea Sector of West Antarctica, particularly the Thwaites and Pine Island Glaciers (Rignot et al., 2019). Sediment records from the Amundsen Sea indicate that Thwaites and Pine Island have been in disequilibrium since at least the 1940s **TODO: cite**. Direct measurements of Pine Island Glacier made possible by the satellite era reveal a consistent pattern of change: retreat of the glacier terminus, acceleration of its ice stream, and a subsequent thinning as the glacier adjusts to maintain its flux balance (Joughin et al., 2003; Payne et al., 2004) **TODO: cite more recent**. It is difficult to draw direct links between the current disequilibrium and anthropogenic climate change, but it's likely that climate change has at least exacerbated the retreat of Pine Island glacier over this period (Bradley et al., 2025). **TODO: See Ines's 2023 remote sensing review paper for more inspiration on Pine Island. Also, Davison 2023 paper on ice shelf mass balance could be useful here**.

TODO: Thwaites

Unlike the Arctic, which has warmed by **TODO: value** times the global average due to nonlinear feedbacks (Previdi et al., 2021), Antarctic surface air temperatures have broadly followed the global trend **TODO: cite**. The one exception to this is the Antarctic Peninsula, which has warmed three times faster than the rest of the continent, up to temperatures otherwise unprecedented in the Holocene (Vaughan et al., 2003). This atmospheric warming has increased surface melting of the Antarctic Peninsula Ice Sheet and its resulting sea level contribution (Vaughan, 2006). The most pronounced changes in the APIS have been the thinning, retreat, and collapse of some of the peninsula's ice shelves (Cook and Vaughan, 2010; Rignot et al., 2013). The extent of ice shelves on the peninsula roughly trace the -9°C isotherm, suggesting a "limit of viability" for APIS ice shelves that will shrink with further warming (Morris and Vaughan, 2013). Although this is a useful heuristic, the mechanisms of ice shelf collapse are not purely thermal (see, for example, Sections 1.3.1 and **??**).

The first ice shelf collapse to occur during the satellite era was the Wordie Ice Shelf in 1989 (Doake and Vaughan, 1991). This was followed by a sequential collapse from North to South down the Eastern flank of the peninsula, starting with the Larsen inlet — also in 1989 — then the Prince Gustav and

Larsen A ice shelves in 1995 (Cook and Vaughan, 2010). Finally, the Larsen B ice shelf progressively thinned between 1992 and 2001 before its eventual collapse over a few days in February 2002 (Shepherd et al., 2003). The Larsen C ice shelf has so far avoided collapse, but has continued to thin at a rate of -3.8 m per decade over increasingly large areas of the ice shelf (Paolo et al., 2015). Between August 2015 and July 2017, a 200 km crack developed along the ice shelf, leading to the calving of a large iceberg representing 10% of the total ice shelf area (Hogg and Gudmundsson, 2017). Due to the loss of buttressing force provided by ice shelves (see Section 1.3.1), the collapse of each ice shelf on the peninsula was consistently followed by an acceleration and thinning of its upstream glaciers (Rignot et al., 2004; Glasser et al., 2011; Dømgaard et al., 2025). The collapse of the Prince Gustav and Larsen A ice shelves cannot be uniquely attributed to anthropogenic climate change, since sediment records indicate that they have previously collapsed during the mid-Holocene (Pudsey and Evans, 2001; Brachfeld et al., 2003). This is corroborated by written accounts of the Prince Gustav ice shelf by 19th century polar explorers suggesting that it had been in retreat since at least 1843 (Cooper, 1997). However, similar records from ocean sediments previously beneath the Larsen B ice shelf indicate that it had been stable throughout the Holocene, and therefore suggest that its recent collapse contains an anthropogenic signal (Domack et al., 2005).

NOTE: Too much on the Peninsula? Two big paragraphs on the smallest region (and least important for my work) seems like a lot. On the other hand, the peninsula is an excellent case study for what might happen elsewhere in Antarctica, an argument which I plan to develop in later sections, so it is perhaps worth going into detail here.

In contrast to the WAIS and APIS, the East Antarctic Ice Sheet remains close to a state of balance, with a slight positive mass trend of 3 ± 15 Gt yr^{-1} between 1992–2020 (Otosaka et al., 2023). The high uncertainty in this estimate is the result of disagreement between different methods of measurement, with some studies showing a negative mass trend over similar periods (e.g. Rignot et al. 2019; Shepherd et al. 2019). These aggregate measurements also belie the spatial variability in East Antarctic mass trends. Several studies highlight regional mass loss from Wilkes Land in East Antarctica, particularly the Totten Glacier, which drains the Aurora subglacial basin (Shepherd et al., 2019; Velicogna et al., 2020; Smith et al., 2020). This is paired with a positive mass trend over the East Antarctic plateau, consistent with reanalysis-derived increases in precipitation (Davis et al., 2005). This spatial variability suggest that whilst the East Antarctic Ice Sheet is in a state of balance, it is not in a state of equilibrium, with the potential for more significant mass change if this balance is disrupted.

TODO: Check Jamie's and Mira's theses to look for more EAIS notes.

TODO: Quick synthesis of this section?

1.2.3 Antarctica in past climates

972 words

NOTE: Comparing this to Jamie's section on Antarctica in the Pliocene, mine is much more condensed and missing quite a lot of context. However, I wouldn't expect my section to be as big as Jamie's anyway, given the relative importance of the Pliocene in my thesis. For example, I essentially pick up with Pliocene sea level estimates in 2015, whereas Jamie gives a longer history. I should probably include some discussion of palaeoshoreline work, which gave even higher sea level estimates than d18O. I could also spend time explaining how the δ^{18}O record works. I

generally feel that I am rushing through references without giving them proper explanation or room to breath, but this may also be a matter of taste.

Past climates are a natural laboratory for exploring the how the Earth system responds to changes in greenhouse gases. By comparison with the modern era, they can inform projections of the future climate (Tierney et al., 2020). The Mid-Pliocene Warm Period (mPWP), a period 3.3–3.0 million years ago with CO_2 concentrations near 400 ppm and temperatures 3–5.3°C warmer than present (Tierney et al., 2025), is the closest analogue to end of the century projections under medium to high emissions (Burke et al., 2018). **NOTE: I've used the Tierney 2025 numbers here because they seem quite convincing, but perhaps I should stick to the IPCC values and simply mention that it could be warmer?** Early attempts to estimate Pliocene sea level based on the the δ^{18}O record — preserved in the layers of benthic foraminifera that occupy ocean sediment cores — produced highly uncertain results (Dutton et al., 2015). For example, Miller et al. (2012) estimated that sea level was 22 ± 10 m higher than today, the lower bound of which would require the near total loss of the Greenland and West Antarctic Ice Sheets **NOTE: This is just technically false because miller et al use a multiproxy approach, not just d18O**. The upper range of these estimates then requires significant contribution from marine and possibly terrestrial sectors of East Antarctica. Much lower sea level estimates can be obtained by using a higher δ^{18}O of Antarctic ice, justifiable by warmer temperatures of the Pliocene (Winnick and Caves, 2015; Gasson et al., 2016), but the true value is then highly uncertain. Raymo et al. (2018) highlight further biases in the δ^{18}O record that could both under- and over-estimate Pliocene sea level.

NOTE: I've again condensed a lot of information below compared to

Jamie's. The references I've cited could all be explained in much more detail, but perhaps this level of detail is appropriate for my thesis?

Regional palaeoclimate records provide a suite of evidence for a dynamic WAIS that advanced and retreated in obliquity-paced glacial-interglacial cycles: Naish et al. (2009) and Seidenstein et al. (2024) found open-ocean diatom species and warm-water foraminifera in sediments beneath the current Ross ice shelf; Gohl et al. (2013, 2021) reveal buried grounding zone wedges in various locations of the stratigraphic record; and Passchier et al. (2025) and Horikawa et al. (2026) demonstrate a cyclicity in the sediment composition of the continental shelf. In contrast, evidence for change in the EAIS is more fragmentary, but an increasing number of palaeoclimate records point to periods of substantial retreat. Sediment cores offshore of the George V and Adélie coasts show a sequence of change in ocean productivity, ice-rafted debris, and sediment provenance pointing to retreat in the Wilkes subglacial basin (Cook et al., 2013; Patterson et al., 2014; Bertram et al., 2018). Ice rafted debris from Wilkes land and Adélie land has been found as far away as Prydz Bay, which would require large icebergs to transport, perhaps comparable to Northern Hemisphere Heinrich events (Williams et al., 2010). This is supported by deep glacial erosion representing a plausible location for the terminus of Totten Glacier in the Pliocene, 150 km behind its current grounding line (Aitken et al., 2016). Conversely, seismic soundings of the Aurora subglacial basin evidence the existence of ice near current grounding lines since the late Miocene (Gulick et al., 2017). The EAIS also appears to have responded differently to orbital forcing than the WAIS, lacking a clear obliquity signal in its climate records (Patterson et al., 2026).

NOTE: Again, the below paragraph could go into much more detail, particularly about PLISMIP. I'm interested to know if you feel any particu-

lar paragraph in this section is too short, or whether they are all too short (or maybe not.)

The regional records provide coherent evidence of significant mass loss from the WAIS and at least some of the marine sectors of the EAIS. However, ice sheet models in the Pliocene Ice Sheet Model Intercomparison Project (Dolan et al., 2012) were unable to reproduce this mass loss when forced by a Pliocene climate (De Boer et al., 2015; Dolan et al., 2018). Most (but not all) models simulated a collapse of the WAIS, but this was often offset or even exceeded by accumulation over East Antarctica. The theory of marine ice cliff instability (Section 1.3.2) was first popularised by its ability to dramatically increase Antarctic sea level contribution in the mPWP (Pollard et al., 2015). However, it was not required to reach the lower range of Pliocene sea level when accounting for modelling and reconstruction error (Edwards et al., 2019). Other models were later able to simulate higher mass loss from Antarctica without MICI, including from the marine sectors of East Antarctica (Golledge et al., 2017b; Blasco et al., 2024). Furthermore, a grain size proxy that avoids the high uncertainties related to δ^{18}O suggests that sea level oscillated by 4.1–20.7 m over Pliocene glacial-interglacial cycles (Grant et al., 2019; Grant and Naish, 2021). If taken as a deviation from present-day sea level, the upper bound would still require mass loss from marine sectors of East Antarctica, but not from terrestrial sectors.

This progress in both models and palaeoclimate records has made significant progress towards closing the gap between them. Both support a dynamic WAIS that collapsed in climates not so different from that projected by the end of this century. Marine sectors of the EAIS may also be vulnerable to anthropogenic warming, but the palaeoclimate records cannot tell us anything about the pace of retreat — whether it would occur on multi-century timescales. The uncertainties in sea level reconstructions are still very high, but the changes to Antarctica are very large, making them a natural complement to observations in the modern era (with small uncertainties but small changes) when assessing the results of ice sheet models.

1.3 Ice sheet dynamics

1.3.1 Ice-ocean interactions

974 words

Interactions between the ocean and Antarctica's ice shelves are an important driver of ice sheet mass loss (Seroussi et al., 2020; Coulon et al., 2024). Although ice shelves don't contribute directly to sea level rise, they have long been recognised to influence mass loss through the buttressing force they impose on inland ice streams (Dupont and Alley, 2005). Buttressing is mostly driven by the lateral drag imposed at the ice shelf boundary, which is transmitted back to the grounding line as a resistance to outflow. Buttressing strength has been explored in models by measuring the instantaneous change in grounding line flux following changes to — or removal of — the ice shelf. These experiments show that buttressing is sensitive to ice shelf thinning (Reese et al., 2018; Gudmundsson et al., 2019; Greene et al., 2022), calving (Greene et al., 2022), and damage (Lhermitte et al., 2020). Comparison against time-varying observations of calving and ice shelf fracture confirm that these processes are responsible for the subsequent acceleration of upstream ice (Sun and Gudmundsson, 2023; Surawy-Stepney et al., 2023). Buttressing strength tends to be higher for ice shelves in tightly enclosed basins such as the Amery or Moscow University Ice Shelves (Greene et al., 2022). In contrast, the current Thwaites Ice Shelf is largely unenclosed except for a pinning point at its Eastern boundary, which results in comparatively less buttressing (Greene et al., 2022; Gudmundsson et al., 2023). The total loss of all Antarctic ice shelves would raise sea level by several metres on multi-century timescales, although the response varies substantially between models (Sun et al., 2020).

Changes to the geometry of an ice shelf are influenced by its interactions with the ocean — melting or refreezing occurs at the ice-ocean interface depending on the in-situ temperature and salinity of seawater. These vary significantly around Antarctica and are greatly influenced by large-scale ocean circulation and the circulation within ice shelf cavities (Wang et al., 2023). Of particular importance is the role of Circumpolar Deep Water (CDW), a warm, salty water mass, which occupies intermediate depths of the Southern Ocean (Wang et al., 2025). CDW shoals further South due to upwelling by the prevailing winds — where the mid-latitude Westerlies and polar Easterlies converge, Ekman transport drives a surface divergence that draws up deeper waters (Marshall and Speer, 2012). Here, CDW sits below a cold, fresh layer of Winter Water — the bottom layer and temperature minimum of Antarctic Surface Water (AASW) — delineated by a steep temperature and salinity gradient that is detectable in ocean depth profiles (Wang et al., 2025). The melting of Antarctic ice shelves is largely a question of whether this CDW is able to access the continental shelf, mix with or displace the cooler waters in ice shelf cavities, and flow down towards the base of the ice sheet. The factors that govern this transport are somewhat outside the scope of this thesis, but we refer the reader to Wang et al. (2023) for a review of the recent literature.

The presence of certain water masses divide ice shelf cavities into broad categories, dominated by different ‘modes’ of melting (Wang et al., 2023). *Warm* cavities are predominantly found in the Amundsen and Bellingshausen Seas. Here, CDW dominates the cavity, driving high region-averaged melt rates of 5–9 m yr^{-1} (Adusumilli et al., 2020). Conversely, the Filchner-Ronne and Ross Ice Shelves both have *cold* cavities with low melt rates and extensive areas of refreezing (Adusumilli et al., 2020). In these cavities, High Salinity Shelf Water (HSSW) is formed through brine rejection during sea ice formation and sinks to occupy the deepest parts of the continental shelf, where it acts as a barrier to CDW. Sea ice formation is enhanced by coastal polynyas found primarily in the Weddell and Ross Seas, the Adelie coast, and Prydz Bay, strengthening the barrier to CDW in their nearby ice shelf cavities (Silvano et al., 2023). HSSW flows under gravity to the grounding line, where it drives comparatively lower and more localised melt than can be found in warm cavities. Melting also occurs near to the calving front where seasonally warmer AASW has been subducted beneath the ice shelf by the easterly winds (Wang et al., 2023). Cold cavities can exhibit tipping point behaviour under climate warming — a reduction in sea ice formation and enhanced meltwater fluxes can weaken the presence of HSSW, allowing CDW to flush the cavity and change it to a warm state (Hellmer et al., 2012, 2017; Hoffman et al., 2024). Hill et al. (2024) project that tipping in the Filchner-Ronne and Ross ice shelf cavities would increase the rate of sea level rise from these sectors, responding similarly to glaciers in the present-day Amundsen Sea.

Over the 20th and 21st centuries, climate change has strengthened the mid-latitude Westerlies and shifted them polewards (Goyal et al., 2021), enhancing the upwelling of CDW (Spence et al., 2014; Wang et al., 2025). Over the Amundsen Sea, changes to the local wind regime have increased the supply of heat to the continental shelf and accelerated the melt of its ice shelves (Holland et al., 2022; Naughten et al., 2022). Thwaites and Pine Island Ice Shelves in particular have respectively lost 70% and 34% of their mass between 1997–2021 through a combination of calving and basal melting (Davison et al., 2023).

The ice shelves have increasingly fractured and fragmented, corresponding with a speedup of their upstream ice (Lhermitte et al., 2020; Joughin et al., 2021; Sun and Gudmundsson, 2023; Surawy-Stepney et al., 2023). The recent mass loss from the Amundsen Sea sector has been widely attributed to these changes (e.g. Gudmundsson et al. 2019; Bett et al. 2026). The remaining ice shelf at Thwaites Glacier is suggested to impose only limited buttressing force on the upstream ice, implying that further ice shelf weakening would have little effect on its contribution to future sea level rise (Gudmundsson et al., 2023).

Outside the Amundsen Sea ice shelf cavities, historical trends in water masses and their links to anthropogenic forcing are less well explored. Several studies have noted the presence of CDW currently circulating through the Totten Ice Shelf cavity (Rintoul et al., 2016; Hirano et al., 2023). Melt rates under the Totten Ice Shelf are currently very high (Adusumilli et al., 2020), but their role in the current Totten Glacier mass trend is unclear (Miles et al., 2025). Nonetheless, this is often cited as a possible driver of dynamic EAIS mass loss in the future (Stokes et al., 2022). Projections of the Antarctic climate, including the Southern Ocean and Antarctic ice shelf cavities, are discussed in more detail in section 1.4.1.

NOTE: Considering moving the 21st century ocean modelling projections from the Climate Context subsection of the Ice Sheet Projections section and moving it here. It would flow into it nicely and the ocean modelling isn't super relevant to the ice sheet projections anyway because they don't use high-res ocean model forcings.

1.3.2 Ice sheet instability

Since the 1970s, it has been suggested that West Antarctica may undergo rapid deglaciation in a warming climate due to instabilities inherent to marine ice sheets (Weertman, 1974; Mercer, 1978). A 1-D flow line model by Schoof (2007) shows that a marine ice sheet overlying a retrograde slope — that is, a slope that deepens towards the centre — has no steady state solution and will undergo self-sustaining retreat. This has become known as Marine Ice Sheet Instability (MISI) and has caused widespread concern that the WAIS may enter a phase of irreversible collapse (e.g. Armstrong McKay et al. 2022). Although it is considered less susceptible in the immediate future, the same collapse mechanisms have also been proposed for marine sectors of the EAIS (Stokes et al., 2022). However, MISI as originally formulated relies on idealised assumptions that are still contested (Sergienko and Wingham, 2022; Sergienko and Haseloff, 2023). Realistic ice sheets can be stabilised by pinning points in the bed topography (Favier et al., 2012, 2016), ice shelf buttressing (Pegler, 2018; Cornford et al., 2020), or surface accumulation feedbacks (Sergienko, 2022). Furthermore, many studies don't distinguish the dynamic MISI response from a second feedback whereby ungrounding increases the area over which melt is applied (Arthern and Williams, 2017). This can make the role of MISI in governing past, present, and future ice loss difficult to isolate.

The fundamental difficulty with identifying MISI in model simulations is disentangling the instability response from the persistent forcing. Experiments that show continued acceleration of mass loss despite a *reduction* in forcing therefore tend to offer the most convincing evidence (e.g. Feldmann and Levermann 2015; Gandy et al. 2018). Another line of evidence is to demonstrate hysteresis of the ice sheet or glacier, which is symptomatic of a bistable system (Garbe et al., 2020; Rosier et al., 2021; Reed et al., 2024). Several studies have focused on detecting MISI in the recent retreat of Pine Island Glacier, highlighting the non-linear response following perturbation from its 1970s grounding position (Favier et al., 2014; Reed et al., 2024) and the hysteresis modelled during re-advance (Rosier et al., 2021; Reed et al., 2024). In this case, the glacier finds stability on subglacial ridges in the bedrock, separated by a retrograde slope over which MISI dominates. Rosier et al. (2021) identify two further stabilising ridges behind the current grounding line, after which the glacier collapses completely. Similar studies have been completed for Thwaites glacier, with roughly equivalent conclusions regarding its stability and recent retreat (Joughin et al., 2014; Waibel et al., 2018). Feldmann and Levermann (2015) suggest that the collapse of the Amundsen Sea glaciers would destabilise the entire West Antarctic Ice Sheet, committing it to multi-metre sea level rise. However, this is unlikely to occur this century; rather, multi-century or millennial-scale projections are required in order to fully understand the effects of MISI on future sea level rise (Seroussi et al., 2024; Klose et al., 2024).

There is clear interest in whether MISI is already underway in West Antarctica and whether the current mass trends are reversible. Reese et al. (2023) suggest that current conditions are sufficient to collapse Thwaites Glacier and possibly sufficient to collapse Pine Island Glacier over several thousand years. On the other hand, the same model configurations can be used to show that grounding lines, when perturbed from a steady state near to their present-day location, will re-advance if the forcing perturbation is removed (Hill et al., 2023). This suggests that while the current grounding lines are not unstable, reversing the mass trend in West Antarctica would require at least a return to preindustrial climate conditions — though this is considered unlikely even in the most optimistic projections (Naughten et al., 2023). It is also important to note that the grounding line retreat simulated by Hill et al. (2023) does not cross any of the subglacial ridges identified by Rosier et al. (2021) as thresholds for the irreversible retreat of Pine Island Glacier. The duration and intensity of forcing required to cross these thresholds is not quantified in either study.

In addition to MISI, there is also interest in a much more recently proposed instability related to the failure of tall ice cliffs. Bassis and Walker (2012) show that marine ice cliffs, left behind following the loss of an ice shelf, will collapse under their own weight if they exceed a critical height. This effect is amplified if coupled with melt-induced hydrofracture at the surface (Pollard et al., 2015). Over a retrograde bed, cliff height increases as the calving front retreats, hence leading to self-sustaining collapse. This has become known as Marine Ice Cliff Instability and has since gained significant attention due to its capacity to dramatically increase multi-century sea level projections (DeConto et al., 2021). MICI has been invoked as the driver of sea level change in past climates (DeConto and Pollard, 2016), perhaps evidenced by iceberg-keel plough marks in Pine Island Bay (Wise et al., 2017). However, it has yet to be observed in the modern era, nor is it necessary to achieve past sea level change (Edwards et al., 2019). More recent implementations of MICI into ice sheet models has shown that the calving front is stabilised as ice deformation lowers cliffs below their critical height for collapse (Morlighem et al., 2024; Sun and Gudmundsson, 2026).

1.3.3 GRD effects

On timescales spanning centuries or millennia, interactions between the ice sheet and solid Earth become relevant for Antarctic projections (Whitehouse et al., 2019). The redistribution of mass by an ice sheet influences the Earth's rotation and gravitational field; it also alters the load that it exerts on the Earth's crust and mantle, which respond through visco-elastic deformation. These changes are collectively referred to as GRD effects (Gravity, Rotation, Deformation; Gregory et al. 2019); where they arise from past changes in land ice, they are known as Glacial Isostatic Adjustment (GIA; Gregory et al. 2019), although we note that the literature can sometimes use these terms interchangeably.

GRD effects influence ice dynamics through changes in local sea level, which arise from two sources: gravitational and rotational changes to the geoid, and vertical land movement driven by solid Earth deformation. In the case of an ice sheet that is losing mass, the load that it exerts on the Earth's crust decreases, inducing uplift of the land surface due to isostatic rebound. At the same time, the gravitational pull of the ice sheet is reduced, lowering the geoid in the near-field (and raising it in the far-field). Both of these effects reduce depth at the grounding line of a marine ice sheet, which can mitigate the effects of MISI (Gomez et al., 2010). This coupling between GRD effects and ice sheet dynamics is referred to as 'the sea-level feedback' and generally lowers sea level projections relative to those with a fixed bed and geoid (Gomez et al., 2015; Larour et al., 2019; Gomez et al., 2024; Han et al., 2025).

Sensitivity to the sea-level feedback is highest in regions with retrograde bedrock such as the Amundsen Sea sector (Larour et al., 2019; Han et al., 2025). Studies of Pine Island and Thwaites Glaciers specifically find that bed deformation mitigates mass loss by 30% and 20% respectively (Kachuck et al., 2020; Book et al., 2022). However, the feedback strength is also sensitive to the seismic structure of the solid Earth, with lower mantle viscosity generating faster uplift and higher mitigation of mass loss (Gomez et al., 2015; Coulon et al., 2021). This structure varies spatially, with a generally thick lithosphere and high mantle viscosity underlying East Antarctica, but relatively thin lithosphere and low mantle viscosity under areas of West Antarctica, notably the Amundsen Sea sector (Lloyd et al., 2020). This spatial variability drives different responses in the East and West Antarctic ice sheets — for example, Coulon et al. (2021) find that visco-elastic effects have little influence over the collapse of the Wilkes basin. Models that fail to account for this spatial variability tend to underestimate uplift in West Antarctica or overestimate it in East Antarctica, with knock-on effects for mass loss and sea level rise (Coulon et al., 2021; Gomez et al., 2024; Han et al., 2025).

The sensitivity of ice dynamics to GRD is also controlled by the rate of mass loss, since very rapid mass loss can outpace the visco-elastic response of the solid Earth. This leads to West Antarctic collapse in experiments with very high forcing (Gomez et al., 2015; Coulon et al., 2021; Gomez et al., 2024) or with MICI (Gomez et al., 2024), irrespective of GRD effects. Furthermore, the continued uplift following collapse expels water from Antarctic marine basins, exacerbating far-field sea level rise. In extreme cases, this can flip the feedback such that it is a net-positive contribution to sea level rise (Gomez et al., 2024).

1.4 Projections of the Antarctic ice sheet

1.4.1 Climate context

683 words

Climate projections in AR6 were informed by the Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al. 2016), a coordinated set of climate experiments organised by the World Climate Research Programme. AR6 structured its climate projections as a collection of storylines, each describing a narrative of how society develops over the 21st century (O'Neill et al., 2016). In the previous assessment report (CMIP5; Taylor et al. 2012), projections followed the Representative Concentration Pathways (RCPs; van Vuuren et al. 2011), which were defined only by their greenhouse gas emissions. AR6 developed this by introducing the Shared Socioeconomic Pathways (SSPs; Riahi et al. 2017), which considered the technological development, international collaboration, and socioeconomic conditions that shape our ability to mitigate and adapt to climate change. For example, SSP1 is a sustainable pathway with a focus on human wellbeing, whereas SSP5 is a growth-oriented pathway with high fossil fuel consumption. Integrated Assessment Models then combined these with specific climate policy targets to produce their emissions trajectories. These targets were chosen to roughly align the radiative forcing of the SSPs with the RCPs: SSP1-2.6 with the low-emissions RCP2.6 and SSP5-8.5 with the high-emissions RCP8.5. Between them, they represent the spread in possible emission futures, though SSP5-8.5 is now considered unlikely given the pace of global decarbonisation (Hausfather, 2025). For the purposes of this thesis, SSP5-8.5 represents more of a worst-case ceiling rather than a likely projection.

AR6 projected a late 21st century global climate that is 0.6–2.0 °C warmer than the 1995–2014 climatology in SSP1-2.6 and 2.7–5.7 °C warmer in SSP5-8.5 (Lee et al., 2021). The change to the Antarctic climate is similar but more uncertain, between 0.1–2.9 °C warmer in SSP1-2.6 and 1.7–5.6 °C in SSP5-8.5. It projected continued warming and freshening of Southern Ocean surface and intermediate waters, which increases evaporation at the ocean surface and is linked to higher precipitation over Antarctica (Fox-Kemper et al., 2021; Douville et al., 2021). The models predict an average increase in precipitation of 5.5% per degree of warming, exceeding 10% over the East Antarctic Plateau (Nicola et al., 2023). This increases the total surface mass balance of the Antarctic ice sheet, but is partially offset under high warming by enhanced surface melting of Antarctic ice shelves (Kittel et al., 2021).

AR6 had *low confidence* in projected changes in the supply of Circumpolar Deep Water to Antarctic ice shelves due to numerous limitations of the CMIP6 models: firstly, they did not resolve ice shelf cavities, meaning they could not capture the cavity circulation, mixing, or the bathymetric features that are important for modelling basal melt rates (Burgard et al., 2022); secondly, their resolution was too coarse to resolve the eddy processes that help to transport CDW onto the continental shelf (Wang et al., 2023); thirdly, persistent warm biases in the Southern Ocean across multiple generations of climate models raise concerns about the validity of projected changes (Barthel et al., 2020); and lastly, climate models at the time included only a static ice sheet, ignoring the ice-ocean meltwater feedbacks that can affect the strength of CDW reaching the base of ice shelves (section 1.3.1). Instead, high-resolution ocean models, which resolve ice shelf cavities, eddies, and include meltwater fluxes, are required to confidently project changes in ice shelf basal melt (Yung et al., 2026). Such simulations of the Amundsen Sea project an enhanced transport of CDW onto the continental shelf and accelerated melting of its ice shelves (Jourdain et al., 2022; Naughten et al., 2023). Naughten et al. (2023) show that this is nearly independent of the future emissions scenario, implying that these 21st century changes are now unavoidable. The crossing of a tipping point is also possible this century for the Filchner-Ronne (Teske et al., 2024; Song et al., 2025), Ross (Jacobs et al., 2022; Siahaan et al., 2022), and Amery ice shelves (Jin et al., 2025), though the timing and likelihood of this transition remains uncertain and model-dependent (Naughten et al., 2021; Comeau et al., 2022; Kusahara et al., 2023).

The climate of the late 23rd century is strongly dependent on future emissions, with global temperatures 6.6–14.1 °C higher than the preindustrial era in SSP5-8.5, but only 0.6–1.4 °C higher in SSP1-2.6 (Lee et al., 2021). Under high emissions, the Antarctic climate would be radically different than today, with surface melt possibly overtaking snow accumulation to drive an overall negative surface mass balance (Coulon et al., 2024). The Southern Ocean could become virtually free of sea ice, with CDW occupying most ice shelf cavities and basal melt rates increasing more than 10-fold (Mathiot and Jourdain, 2023). It has been suggested that Antarctica could undergo "Greenlandification" (Mottram et al., 2025), mimicking the conditions that have accelerated Greenland Ice Sheet mass loss in the present day.

1.4.2 21st century projections

1018 words

The Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6; Nowicki et al. 2016) was a CMIP6-endorsed experiment that specifically explored changes in the ice sheets and their contribution to sea level rise. ISMIP6 brought together multiple ice sheet modelling communities to conduct experiments under a consistent protocol. This included experiments exploring the sensitivity of Greenland and Antarctic mass loss to the ice sheet initial states (InitMIP; Goelzer et al. 2018; Seroussi et al. 2019), an experiment exploring the sensitivity of Antarctic mass loss to the sudden collapse of its ice shelves (ABUMIP; Sun et al. 2020), and experiments exploring the Greenland and Antarctic Ice Sheet response to a range of 21st century emission scenarios (Payne et al., 2021). At the time of ISMIP6, the climate model outputs from CMIP6 were not yet available, so ice sheet models instead used CMIP5 forcing for a range of RCPs. Far-field ocean properties were extrapolated into ice shelf cavities and into regions of grounded ice according to Jourdain et al. (2020). Modelling groups were asked to run a control simulation under a fixed baseline climate, which would be subtracted from the main sea level projections. They could also optionally include the collapse of Antarctic ice shelves, prescribed based on the rate of surface melting, although 21st century collapse was mostly limited to the Antarctic Peninsula and Totten Glacier (Seroussi et al., 2020).

The results of ISMIP6 yielded a Antarctic sea level contribution of between

-1.4–15.5 cm in RCP2.6 and -7.8–30 cm in RCP8.5, with ice shelf collapse increasing the maximum sea level contribution by 2.8 cm (Seroussi et al., 2020). Most of this mass loss originates from the West Antarctic Ice Sheet, with limited mass gain over East Antarctica (Seroussi et al., 2020). However, the highest values were obtained using the most sensitive climate model and basal melt calibration, which were outliers relative to the rest of the ensemble (Edwards et al., 2021). In addition, the control experiment yielded sea level contribution between -13–14.2 cm (Seroussi et al., 2020). This spread is similar to that of the RCPs, underlining the strong influence of non-forcing factors such as initial state and model structural uncertainty, also seen in InitMIP (Seroussi et al., 2019). Edwards et al. (2021) fit a Gaussian process emulator to the ISMIP6 ensemble, using changes in global mean temperature (among other inputs) as training data. This allowed them to combine the ISMIP6 ensemble with the SSPs (via the FaIR simple climate model) to produce probability distributions of future sea level. Emulation has the added benefit of considering information from the whole ensemble and not just the extremes, which considerably narrows the high-end projections (Edwards et al., 2021). This removes the scenario-dependence, with 5–95% confidence intervals of -1–19 cm in SSP1-2.6 and 0–18 cm in SSP5-8.5 (Fox-Kemper et al., 2021). **NOTE: Why do the AR6 numbers here not match the values from the original edwards2021 paper?**

The Linear Antarctic Response Model Intercomparison Project (LARMIP-2), took a different approach to 21st century sea level projections, using linear response theory to estimate the Antarctic sea level response to ocean warming. Unlike the process-based modelling of ISMIP, they conducted a series of idealised experiments which could then be statistically mapped onto the RCPs (Levermann et al., 2020) or SSPs (Fox-Kemper et al., 2021) through changes in global mean temperature. This simpler protocol resulted in a higher number of participating ice sheet models, but it also ignored surface mass balance changes or non-linear effects such as MISI. Even adjusting for surface mass balance, LARMIP-2 obtained a higher 5–95% range than the emulated ISMIP6 ensemble: 3–41 cm in SSP1-2.6 and 1–57 cm in SSP5-8.5 (Fox-Kemper et al., 2021). This was largely due to the high sensitivity of ice shelf melt in LARMIP-2, which corresponds well with melt rates in the Amundsen Sea, but overestimates melt in most other ice shelf cavities (Reese et al., 2020). Nonetheless, AR6 did not make any judgement on whether the ISMIP6 or LARMIP-2 projections were more realistic, conservatively combining LARMIP-2 with the emulated ISMIP6 to yield a 5–95% confidence interval of -1–41 cm in SSP1-2.6 and 0–57 cm in SSP5-8.5 (Fox-Kemper et al., 2021).

A common theme among projections of the Antarctic ice sheet is the competition between the ocean and atmospheric drivers of mass change, the balance of which can determine the sign of Antarctica's sea level contribution (Seroussi et al., 2020; Edwards et al., 2021). These competing effects are manifest in the different responses of the West and East Antarctic Ice Sheets, reflecting the different processes that govern these regions. The marine WAIS is highly influenced by its ice shelves and its response is therefore dominated by ocean warming, whereas the EAIS has a mixed responses of mass loss from its marine sectors and mass gain over the East Antarctic Plateau (Seroussi et al., 2020, 2023; O'Neill et al., 2025). Uncertainty in the ensemble is overall dominated by the choice of ice sheet model, but drivers of uncertainty can vary even on a glacier-scale (Seroussi et al., 2023).

Since AR6, several studies have given more context to the results of ISMIP6 and LARMIP-2. For example, the linear basal melt parameterisation used in LARMIP-2 has been shown to perform poorly relative to the quadratic parameterisation used in ISMIP6 or other basal melt parameterisations (Burgard et al., 2022). On the other hand, the basal melt sensitivity used for high-end projections in ISMIP6 compares poorly with coupled ocean-ice shelf models, suggesting that these were overestimated (Jourdain et al., 2022). Improvements have been made to ice sheet or climate models, including several climate models that have implemented a coupled dynamic ice sheet (Muntjewerf et al., 2021; Siahaan et al., 2022; Park et al., 2023; Goelzer et al., 2025; Sadai et al., 2025). Only a few of these have made 21st century projections with a coupled Antarctic Ice Sheet, but they highlight the complex meltwater feedbacks that can both enhance and inhibit mass loss (Park et al., 2023). Overall, these feedbacks tend to reduce Antarctica's mass loss (Park et al., 2023; Sadai et al., 2025) or even yield a mass gain (Siahaan et al., 2022), but the limited number of these studies means that this balance is still uncertain and it may change on longer timescales.

An important conclusion of AR6 was that the Antarctic Ice Sheet is *likely* to continue losing mass throughout this century, irrespective of future emissions scenario (Fox-Kemper et al., 2021). It emphasised that whilst the scenarios with higher warming drive a different response in the ice sheet, there was no consensus on which of the competing effects driving mass loss or mass gain would emerge as dominant over the 21st century. This makes the influence of 21st century emissions over Antarctic mass loss difficult to assess, demonstrating the need for longer-timescale projections if we are to fully understand the consequences of present-day climate policy on future sea level rise (Lowry et al., 2021).

1.4.3 Multi-century projections

783 words

As explained in the introduction to this chapter, ISMIP6 did not make projections of sea level contribution beyond 2100, with multi-century projections relying on extrapolation, expert elicitation, and disparate modelling studies to make the AR6 assessment. This is largely due to a lack of appropriate forcing, since state of the art climate model simulations did not generally extend past 2100 at this time. The RCPs were extended to 2500 as described in Meinshausen et al. (2011). However, multi-century projections in CMIP5 were reliant on models of intermediate complexity (Zickfeld et al. 2013). These were less sophisticated than the more complex General Circulation Models (GCMs) and do not output the necessary climate variables to run an ice sheet model. Multi-century ice sheet projections then either fixed the climate after 2100 (Bindschadler et al., 2013; Lowry et al., 2021; Lipscomb et al., 2021; Chambers et al., 2022), used an idealised forcing (Martin et al., 2019; Garbe et al., 2020), or constructed an appropriate forcing through statistical methods (Golledge et al., 2015; Bulthuis et al., 2019; DeConto et al., 2021; Greve et al., 2023). For example, Golledge et al. (2015) and Bulthuis et al. (2019) extend the 2100 CMIP5 GCM projections by scaling them linearly with temperature, which is derived from the CMIP5 intermediate complexity models. This provides a proxy for RCP-based multi-century climate simulations, but assumes that the relationship between temperature and other climate variables stays constant with time. Nonetheless, these studies highlight the importance of multi-century forcings, which can double sea level projections relative to a forcing that's fixed after 2100 (Seroussi et al., 2024; Greve et al., 2023).

Since ISMIP6, several climate models (without coupled ice sheets) have now ran climate simulations up to 2300 using extended SSPs (Meinshausen et al., 2020). This has, for the first time, allowed multi-century ice sheet projections without the need for statistical extrapolation of climate forcing. Seroussi et al. (2024) conducted an ISMIP6-style experiment designed for multi-century projections of the Antarctic Ice Sheet. The experiment focused mainly on a high emissions future scenario, with only 2 out of 14 recommended forcings representing a low emissions pathway. The results, gathered from 16 participating ice flow models, highlight the acceleration of mass loss from Antarctica that occurs after the end of the 21st century, up to 4.4 m sea level equivalent by 2300. Ice shelf collapse affects most regions with floating ice by the end of the 23rd century, increasing sea level contribution by 1.1 m on average. The experiment produced dramatic changes in the WAIS by 2300, with grounding line retreat along the Siple and Weddell coasts likely to exceed 200 km. Nearly half of models initiate a retreat of Thwaites Glacier, finding stability near to a pinning point 400 km behind its current grounding line. The EAIS changes from a net mass gain over the 21st century to a net mass loss by the end of the 23rd century, again originating from its marine sectors. However, this is muted in comparison to studies that include MICI, which leads to significant ungrounding in the Wilkes and Aurora basins (DeConto et al., 2021; Gomez et al., 2024). The response under low emissions is less well explored — the limited simulations by Seroussi et al. (2024) project a sea level contribution between -0.37 and 0.62 m, but other studies suggest that Thwaites Glacier could continue to retreat, raising sea level by up to 2 m (Klose et al., 2024; Coulon et al., 2024, 2025).

Uncertainty in multi-century projections remains very high, but several efforts have been made to characterise and quantify it (Bulthuis et al., 2019; Coulon et al., 2024, 2025). As with 21st century projections, the ocean is the primary driver of overall mass loss throughout the 22nd and 23rd centuries due to its strong control on the WAIS (Bulthuis et al., 2019; Greve et al., 2023; Coulon et al., 2024). However, the East Antarctic ice sheet response is increasingly driven by atmospheric changes after 2100 and can be highly dependent on the forcing climate model (Coulon et al., 2025). Significantly stronger forcing over three centuries can overcome some of the ice sheet model structural uncertainty that dominated 21st century projections (Coulon et al., 2025), but there is still disagreement when considering the full range of ISMIP models (Seroussi et al., 2024). A stronger scenario-dependence also emerges by the end of the 23rd century, with a clean separation of the uncertainties between high and low emission projections (Fox-Kemper et al., 2021; Greve et al., 2023; Coulon et al., 2024; Klose et al., 2024), although wider sampling of uncertainty can still result in a limited overlap (Coulon et al., 2025).

Past the 23rd century, projections are again limited by the available forcing, but long-term simulations highlight the continued mass loss from Antarctica that occurs well after 2300 (Golledge et al., 2015; Bulthuis et al., 2019; Klose et al., 2024; Coulon et al., 2024, 2025). Committed sea level from Antarctica is very sensitive to the forcing climate model, but can reach up to 40 m (Klose et al., 2024). In SSP5-8.5, the WAIS, as well as a significant fraction of the Wilkes basin, can become ungrounded by year 3000 (Bulthuis et al., 2019; Coulon et al., 2024, 2025), possibly followed by all marine sectors of East Antarctica on multi-millennial timescales (Klose et al., 2024). High mitigation can prevent substantial mass loss, but cannot rule out collapse of the Amundsen Sea Sector and committed sea level rise of up to 6 m (Klose et al., 2024). These millennial-scale projections highlight the long response time of ice sheets to high forcing — particularly East Antarctica, which is barely ungrounded by 2300 in models without MICI, but which can experience widespread collapse with no further change in forcing.

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